

习题 6.2

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由 f, g 均为区间 I 上凸函数, 于是由定义可知, 对任意两点 $x_1, x_2 \in I$ 和实数 $\lambda \in (0, 1)$, 总有

$$\begin{aligned}f(\lambda x_1 + (1 - \lambda)x_2) &\leq \lambda f(x_1) + (1 - \lambda)f(x_2) \\g(\lambda x_1 + (1 - \lambda)x_2) &\leq \lambda g(x_1) + (1 - \lambda)g(x_2)\end{aligned}$$

所以

$$\begin{aligned}&\max\{f(\lambda x_1 + (1 - \lambda)x_2), g(\lambda x_1 + (1 - \lambda)x_2)\} \\&\leq \max\{\lambda f(x_1) + (1 - \lambda)f(x_2), \lambda g(x_1) + (1 - \lambda)g(x_2)\} \\&\leq \max\{\lambda \max\{f(x_1), g(x_1)\} + (1 - \lambda) \max\{f(x_2), g(x_2)\}, \lambda g(x_1) + (1 - \lambda)g(x_2)\} \\&= \lambda \max\{f(x_1), g(x_1)\} + (1 - \lambda) \max\{f(x_2), g(x_2)\}\end{aligned}$$

即

$$F(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda F(x_1) + (1 - \lambda)F(x_2)$$

所以, $F(x)$ 也是 I 上凸函数。

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考虑函数 $f(x) = \ln x$ 。

$$\begin{aligned}f'(x) &= \frac{1}{x} \\f''(x) &= -\frac{1}{x^2}\end{aligned}$$

由定理 6.15 可知, f 是 $(0, +\infty)$ 上的凹函数。

由延森 (Jensen) 不定式可知, 对任意 $x_i \in (0, +\infty), \lambda_i > 0 (i = 1, 2, \dots, n)$,
 $\sum_{i=1}^n \lambda_i = 1$, 有

$$\ln(\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n) \geq \lambda_1 \ln x_1 + \lambda_2 \ln x_2 + \dots + \lambda_n \ln x_n$$

(i) 令 $\lambda_i = \frac{1}{n}, x_i = a_i$, 于是

$$\begin{aligned} \ln\left(\frac{1}{n}a_1 + \frac{1}{n}a_2 + \dots + \frac{1}{n}a_n\right) &\geq \frac{1}{n} \ln a_1 + \frac{1}{n} \ln a_2 + \dots + \frac{1}{n} \ln a_n \\ &= \ln a_1^{\frac{1}{n}} a_2^{\frac{1}{n}} \dots a_n^{\frac{1}{n}} \\ &= \ln \sqrt[n]{a_1 a_2 \dots a_n} \end{aligned}$$

由 $\ln$ 在定义域上单增可知,

$$\frac{1}{n}a_1 + \frac{1}{n}a_2 + \dots + \frac{1}{n}a_n \geq \sqrt[n]{a_1 a_2 \dots a_n}$$

即

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}$$

(ii) 令 $\lambda_i = \frac{1}{n}, x_i = \frac{1}{a_i}$, 于是

$$\begin{aligned} \ln\left(\frac{1}{n} \frac{1}{a_1} + \frac{1}{n} \frac{1}{a_2} + \dots + \frac{1}{n} \frac{1}{a_n}\right) &\geq \frac{1}{n} \ln \frac{1}{a_1} + \frac{1}{n} \ln \frac{1}{a_2} + \dots + \frac{1}{n} \ln \frac{1}{a_n} \\ &= \ln \sqrt[n]{\frac{1}{a_1} \frac{1}{a_2} \dots \frac{1}{a_n}} \end{aligned}$$

由 $\ln$ 在定义域上单增可知,

$$\begin{aligned} \frac{1}{n} \frac{1}{a_1} + \frac{1}{n} \frac{1}{a_2} + \dots + \frac{1}{n} \frac{1}{a_n} &\geq \sqrt[n]{\frac{1}{a_1} \frac{1}{a_2} \dots \frac{1}{a_n}} \\ \frac{1}{n} \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) &\geq \frac{1}{\sqrt[n]{a_1 a_2 \dots a_n}} \\ \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} &\leq \frac{n}{\sqrt[n]{a_1 a_2 \dots a_n}} \end{aligned}$$